

PREDICTING HIGH QUANTILES THROUGH THE DIRICHLET PROCESS ON EXTREME MODELLING

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Key words: Beta distribution; Burr distribution; entropy; Pareto distribution; predictive distribution; QQ-plot; tail probabilities; threshold.

Summary: One of the challenging problems in Extreme Value Analysis is that of predicting high quantiles. The Dirichlet process is applied to fitting a model to extreme data exceeding a threshold and the predictive distribution is derived. Through the predictive distribution, a model check using a QQ-plot is provided and high quantiles are predicted. The choice of the threshold is discussed by minimizing the negative differential entropy of the Dirichlet process. The approximation of the tail of a heavy tailed distribution like the Burr by the Pareto or Generalized Pareto and selection of the threshold is investigated by simulation. The prediction of high quantiles is illustrated on a real dataset. We also compare the results with the traditional maximum likelihood approach based on peaks over threshold.

1. Introduction

Let $x_1, \dots, x_k$ denote k ordered observations exceeding the threshold t in a data set of n observations containing extremes. Popular models to fit to such a data set are the Strict Pareto (SP) and the Generalized Pareto (GP) distributions and the method is referred to as the Peak over Threshold (POT) (Smith, 1989; Davison and Smith, 1990; Leadbetter, 1991). Let $F(x|t)$ be

MSC2000: 62G32

the true unknown distribution function (df) of the random variable X and let it be described through the Dirichlet Process (DP) (Ferguson, 1973) according to the partitioning of the sample space by the observed data. If we assume, a priori, that the GPD, say, with df $G(x|t)$, is the model to fit, then, a posteriori, $F(x|t)$ is described through the DP with parameters $\beta G(x|t) + (k+1)F_k(x)$ and $\beta + k$, (see Definition 1 in the appendix) denoted by

$$F(x|t) \sim DP(\beta G(x|t) + (k+1)F_k(x), \beta + k), \quad x > t. \quad (1)$$

Here

$$F_k(x) = \frac{1}{k+1} \sum_{i=1}^k \delta_{x_i}(x) \quad (2)$$

denotes the empirical. $G(x|t)$ is the df given by

$$G(x|t) = 1 - \left(\frac{x}{t}\right)^{-1/\gamma}, \quad x > t \quad (3)$$

for the Strict Pareto, or

$$G(x|t) = 1 - \left(1 + \frac{\gamma}{\sigma_t}(x-t)\right)^{-1/\gamma}, \quad x > t \quad (4)$$

for the Generalized Pareto.

The parameter β is referred to as the concentration parameter. From Corollary 2, $\beta = \alpha(k+1)$, $\alpha > 0$. The choice of α (also referred to as a concentration parameter) for a given k is obtained by minimizing the negative differential entropy (NDE) of the Dirichlet distribution. If $Y_1, \dots, Y_{k+1} \sim \text{Dirichlet}(\nu_1, \dots, \nu_{k+1})$, then the negative differential entropy

of the Dirichlet distribution is given by (Honkela, 2001)

$$\begin{aligned} J_{k+1} &= E(\log(f(Y_1, \dots, Y_{k+1}))) \\ &= -\log \frac{\prod_{i=1}^{k+1} \Gamma(v_i)}{\Gamma(\sum_{i=1}^{k+1} v_i)} + \sum_{i=1}^{k+1} (v_i - 1) (\Psi(v_i) - \Psi(v_0)) \end{aligned} \quad (5)$$

where f denotes the Dirichlet density function and Ψ the digamma function. From Theorem 1 in the Appendix, it follows that J_{k+1} is convex in k and the minimum is reached when $\nu_i = 1$ for all i . This gives a lower bound of $J_{k+1} = \log(\Gamma(k+1))$ which is also convex. The value of k will also be selected through consideration of the NDE. This will be discussed in section 3.

2. Predictive distribution

From the DP follows that marginally $F(x|t, \text{data}) \sim \text{Beta}(a_k, b_k)$ where $a_k = \beta G(x|t) + (k+1)F_k(x)$ and $b_k = \beta + k - a_k$. The mean of the $\text{Beta}(a_k, b_k)$ is given by $\frac{a_k}{a_k + b_k}$ which is also the predictive density of a future observation. Formally, the predictive df of a future X_0 is given by

$$\begin{aligned} H(x_0|t, \text{data}) &= P(X_0 < x_0|t, \text{data}) \\ &= \int_0^1 \pi(u|t, \text{data}) P(X_0 < x_0|u) du \end{aligned}$$

where $\pi(u|t, \text{data})$ denotes the posterior density of $U = F(x|t)$ given the data $x_1, \dots, x_k$, namely $\text{Beta}(a_k, b_k)$ and $P(X_0 < x_0|u) = u$. Therefore

$$\begin{aligned} H(x_0|t, \text{data}) &= \frac{\Gamma(a_k + b_k)}{\Gamma(a_k) \Gamma(b_k)} \int_0^1 h^{a_k} (1-u)^{b_k-1} du \\ &= \frac{a_k}{a_k + b_k} \\ &= \frac{\beta G(x_0|t) + (k+1) F_k(x_0)}{\beta + k}. \end{aligned} \quad (6)$$

The predictive df is a weighted mean between the proposed model $G(x_0|t)$ and the empirical df $F_k(x_0)$. Figure 1 shows schematically the three df's H , G and F_k with the predicted value x_0 corresponding to the p -th quantile.

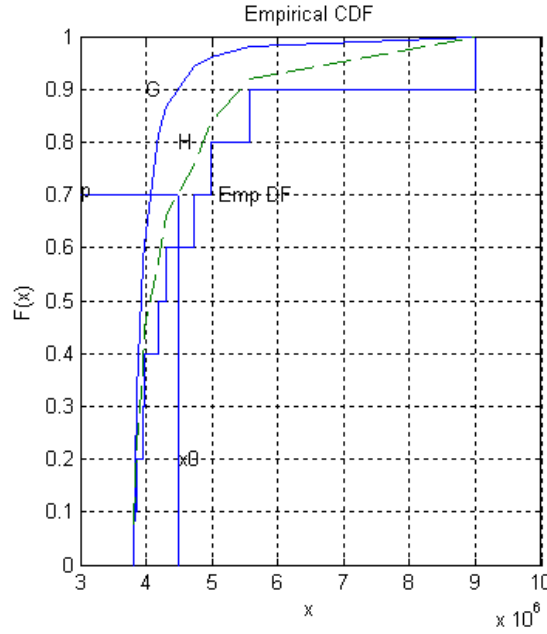


Figure 1. Proposed model $G(x|t)$, empirical df $F_k(x)$ and predictive df $H(x|t, \text{data})$

Let $p = \frac{i}{k+1}$, $i = 1, \dots, k$, then k future quantiles $Q(p)$ can be predicted and compared with the observed quantiles through a QQ -plot. This also served as a measure of goodness of fit of the proposed model $G(x|t)$. The calculation of the predictive quantiles from $H(x|t, \text{data})$ is done numerically by letting $H = p, 0 < p < 1$, and solving for x_0 , the p -th quantile. x_0 is calculated through discretization of the sample space into small intervals and

by choosing a value for p according to these intervals. The parameters of $G(x|t)$ will be estimated in practice.

3. Simulation

Suppose the true df $F(x)$ is $\text{Burr}(\tau, \lambda, \xi)$ with survival function

$$\bar{F}(x) = (1 + \xi \lambda x^\tau)^{-1/\xi}, \quad x > 0, \quad (7)$$

which is perhaps better known in the literature in the form $P(X > x) = \left(\frac{\eta}{\eta + x^\tau}\right)^\lambda$ (see also Theorem 3 in Appendix). The expression (7) shows some relationship with that of the generalized Pareto and we will stick to this notation.

Let $\xi = 1.2920$, $\tau = 4.3067$, $\lambda = 1.6358 * 10^{-28}$ (these values were chosen to simulate a data set fairly similar to that considered in section 4), then the survival function will be as follows:

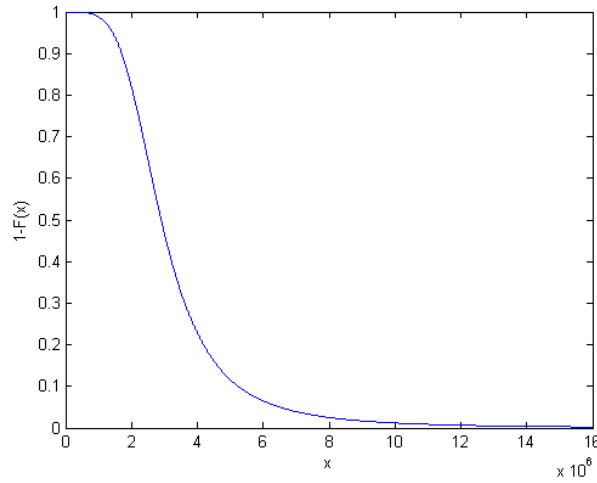


Figure 2. Survival function of the Burr distribution

Illustrating a Strict Pareto fit to the tail of the Burr on simulated data

If we want to fit a Strict Pareto (γ) above a threshold t to the tail, given a random sample of $n = 200$ observations, we assume $G(x|t) = 1 - (x/t)^{-1/\gamma}$, $x > t$, where the Extreme Value Index (EVI) is given by $\gamma = \xi/\tau = 0.3$. This follows fairly easily from

$$\bar{F}(x|t) = \frac{1}{P(X > t)} (1 + \xi \lambda x^\tau)^{-1/\xi} \cong \left(\frac{x}{t}\right)^{-\tau/\xi}, \quad x > t, \quad (8)$$

for $t \rightarrow \infty$.

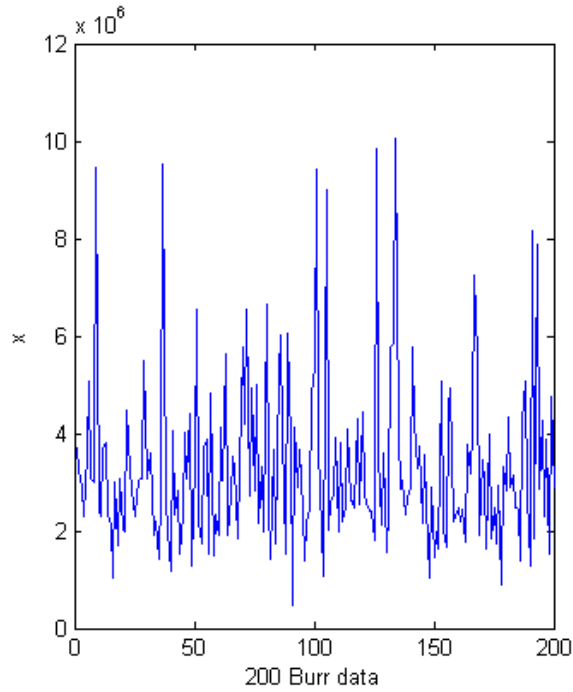


Figure 3. Simulated data from the Burr

Suppose we take the threshold at the largest $k = 108$ observations with $t = 2.7162 \times 10^6$ and taking $\gamma = 0.3$ as known, the NDE J_{k+1} , calculated for different values of α given the partition of the sample space according to the largest k observations shows the convex form in figure 4. The minimum is obtained at $\alpha = 0.153$ which implies maximal information from the data. For $k = 106$, the minimum is obtained at $\alpha = 0.151$. We notice that the NDE is also reduced.

If we repeat this for different k and plot α against k , which corresponds to $\min(\text{NDE})$, we can obtain the number k where the information from the largest k observations gives maximum information on the Pareto fit to the tail.

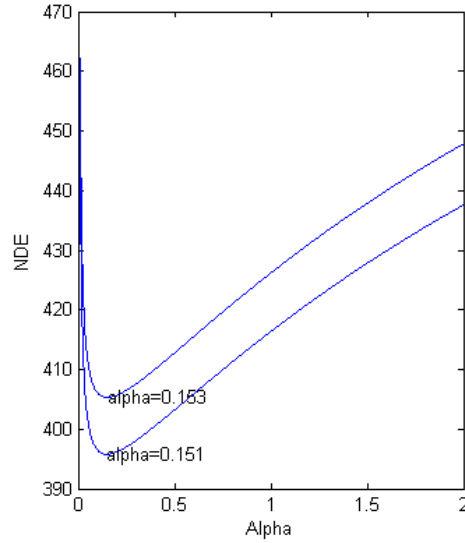


Figure 4. The NDE for different values of α given $k = 108$ (top graph) and $k = 106$

The selected value for α is therefore 0.153 if $k = 108$. If we repeat this for different k , the following graph is obtained:

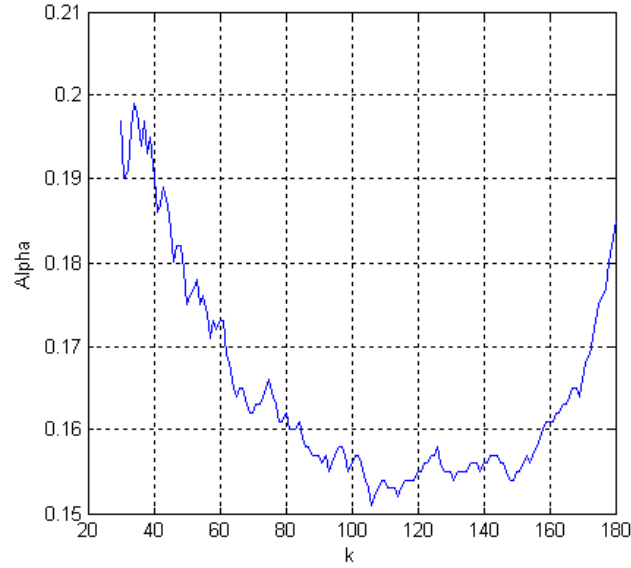


Figure 5. Selected α plotted against different k values

For $k = 106$, the minimum $\alpha = 0.151$ is obtained which is also the value where the NDE reaches a minimum close to the lower bound $\log \Gamma(107) = 391.576$. This is considered to be the optimal value for k where we get maximal information from the data by fitting a Strict Pareto to the tail. Hence the threshold is chosen at $k = 106$ with $t = 2.8419 \times 10^6$.

Figure 6 shows the true conditional tail probabilities from the Burr distribution compared with the conditional probabilities obtained from the

Strict Pareto fitted to tail above the threshold t . The probabilities get closer to each other as x increases.

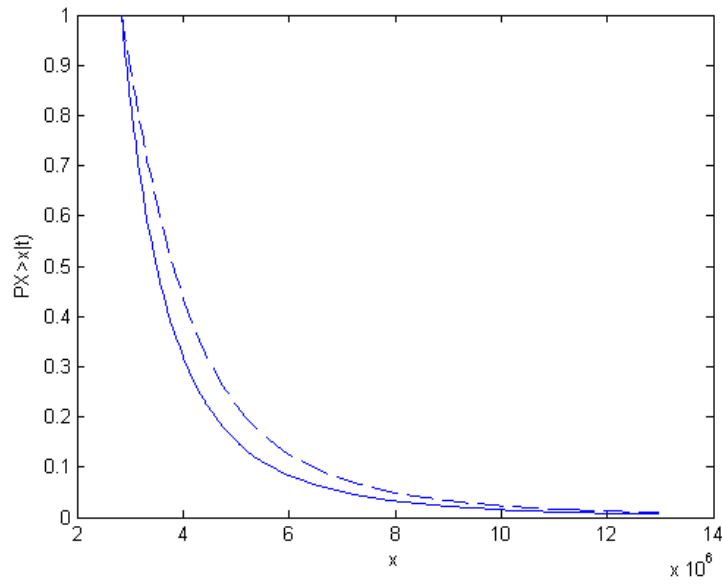


Figure 6. Comparing tail probabilities of the Burr (broken line) with the Strict Pareto (solid line) based in the fit

Taking a large value such as $x = 1.4 \times 10^7$, the true probability $F(x) = P(X < x) = 0.9944$. The unconditional probability according to the Pareto ($\gamma = 0.3$) is $G(x) = 1 - (1 - G(x|t))P(X > t) = 0.9971$ where $P(X > t) = N_t/n = 106/200$. The unconditional predicted probability is $H(x) = 0.9994$. The largest observation was 1.3923×10^7 .

In practice however, γ must be estimated. Using the posterior mean (or mode) of γ as estimates of the tail index γ for different k values, figure 7(a) shows the behaviour. These estimates stay fairly constant close to 0.3 for

k lying between 40 and 75. The MDI prior of α (Beirlant et al., 2004, page 438), namely $\pi(\gamma) \propto \frac{1}{\gamma} \exp(-\gamma)$ is used. Figure 7(b) shows the choice for k taking γ say 0.31 where the relative NDE reaches a minimum. According to this graph, we will select $k = 71$ and the corresponding estimate for γ is $\hat{\gamma} = 0.3212$. The value for α where the NDE reaches a minimum for $k = 71$ is $\alpha = 0.1630$. The threshold at $k = 71$ is $t = 3.5359 \times 10^6$.

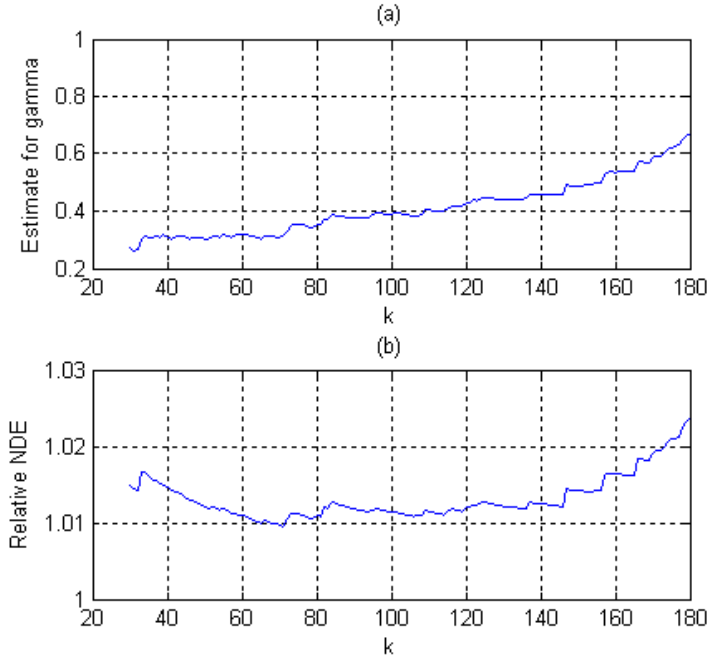


Figure 7. Graphs of (a) $\hat{\gamma}$ and (b) the relative NDE $J_{k+1}/\log \Gamma(k+1)$ against k

A number of simulations from the same Burr distribution similar to the above, have been done and the results turned out to be fairly the same. The k values for choosing the threshold when fitting the Strict Pareto varies between 71 and 156 and the estimates for γ were all just above 0.3. We will now consider the fit of a Generalized Pareto distribution to the tail of the Burr data.

Illustrating the fit of a Generalized Pareto to the tail of the Burr on the simulated data using the true values for the parameters

We will consider the fit of a Generalized Pareto (GP) (4) to the simulated data above shown in Figure 3. The GP is a popular distribution to fit to the tail of Pareto data and therefore we will address the fit. It has two parameters γ (tail index) and σ_t (related to the variation at the threshold t) and we will follow the same process as for the Strict Pareto to find the concentration parameter $\beta = \alpha(k + 1)$ and k . Different models fitted to the tails may require different thresholds. It can be shown (see Appendix, Theorem 3) that for $t \rightarrow \infty$

$$\begin{aligned}\overline{F}(x|t) &= \frac{1}{P(X > t)} (1 + \xi \gamma x^\tau)^{-1/\xi} \\ &\cong \left(1 + \frac{\gamma}{\sigma_t} (x - t)\right)^{-1/\gamma}, \quad x > t\end{aligned}\tag{9}$$

where $\gamma = \xi/\tau$ and $\sigma_t = t\gamma(1 + \xi t^{-\tau})$.

We fitted the Generalized Pareto (8) to the same simulated data above from the Burr($\xi = 1.2920$, $\tau = 4.3067$, $\lambda = 1.6358 \times 10^{-28}$). Figure 8 shows a plot of σ_t against k indicating a minimum at $k = 71$ which coincidentally is also the value where the relative NDE for the Strict Pareto fit reaches its minimum. This will change if the parameters are estimated. From a number of

simulations we did from the Burr and fitting a GP, we found that the minimum of the relative NDE is reached at the same value for k where σ_t (or its estimate) reaches a minimum. The threshold is $t = 3.5359 \times 10^6$. According to figure 9, $\alpha = 0.127$.

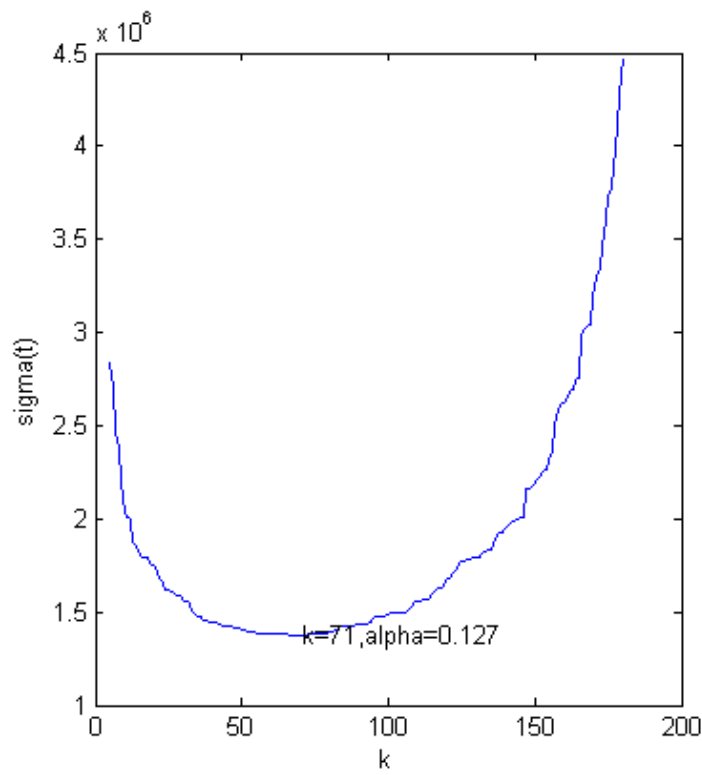


Figure 8. σ_t plotted against different k

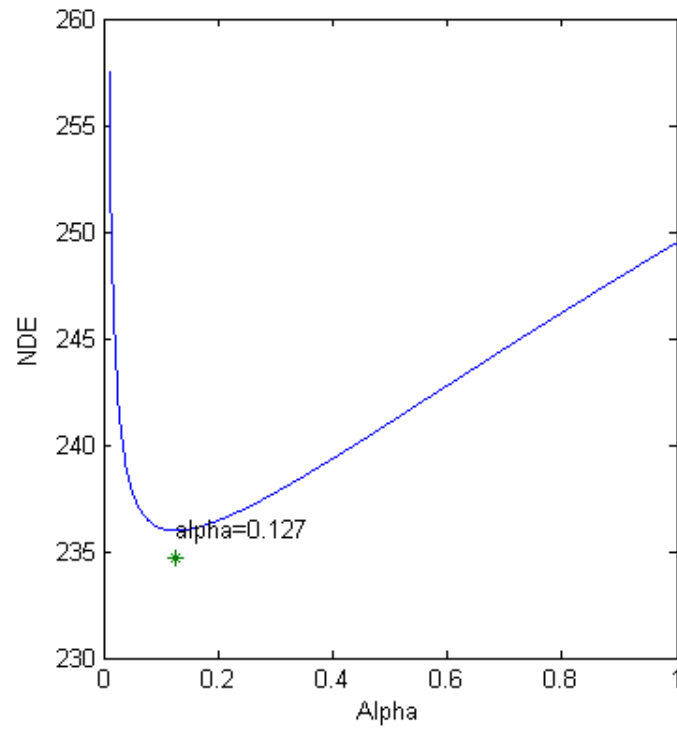


Figure 9. The NDE plotted against different values for α . * indicates the lower bound

The conditional tail probabilities exceeding the threshold t are shown in Figure 10. For comparison, the Strict Pareto fit to the largest $k = 71$ observations is also shown in the graph.

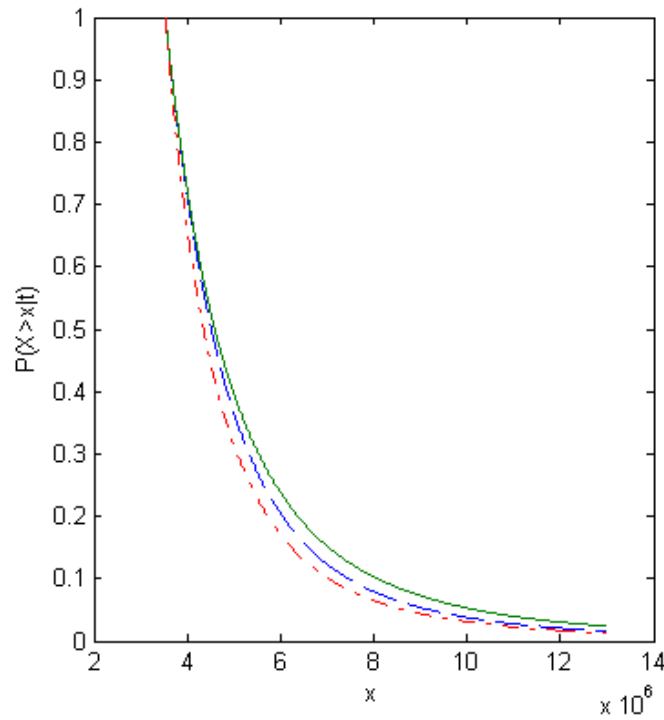


Figure 10. Conditional Burr (broken line), Generalized Pareto (solid line) and Strict Pareto (dotted broken line) tail probabilities exceeding the threshold $t = 3.5359 \times 10^6$

From the graphs we note that the Generalized Pareto over-estimates the true probabilities while the Strict Pareto under-estimates them, but they converge to the true. It seems that the Strict Pareto gives a better fit to the Burr tail than the Generalized Pareto in some areas.

We will not consider estimating the parameters of the GP on the simulated data, but rather continue by fitting the GP to a real data set.

4. Illustration on real data

Beirlant et al. (2004) considered automobile reinsurance claims received by Secura Belgium Re from 1988 until 2001. The data, corrected for inflation, is shown in Figure 11.

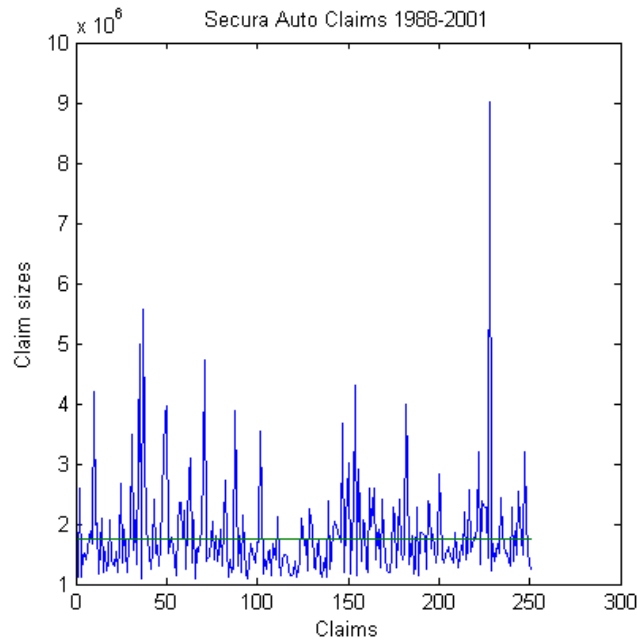


Figure 11. 252 Secura auto insurance claims with threshold $t = 1.7561 \times 10^6$

A GP is fitted to the largest $k = 100$ observations exceeding the threshold $t = 1.7561 \times 10^6$ and the Bayes estimates using MCMC (Beirlant et al., 2004, chapter 11) are given by $\hat{\gamma} = 0.3218$, $\hat{\sigma}_t = 5.4138 \times 10^5$. Figure 12 shows the graphs of the estimates for different values of k and Figure 12(b) shows a

minimum at $k = 100$ ignoring the first part of the graph below $k = 50$ where the estimates vary substantially.

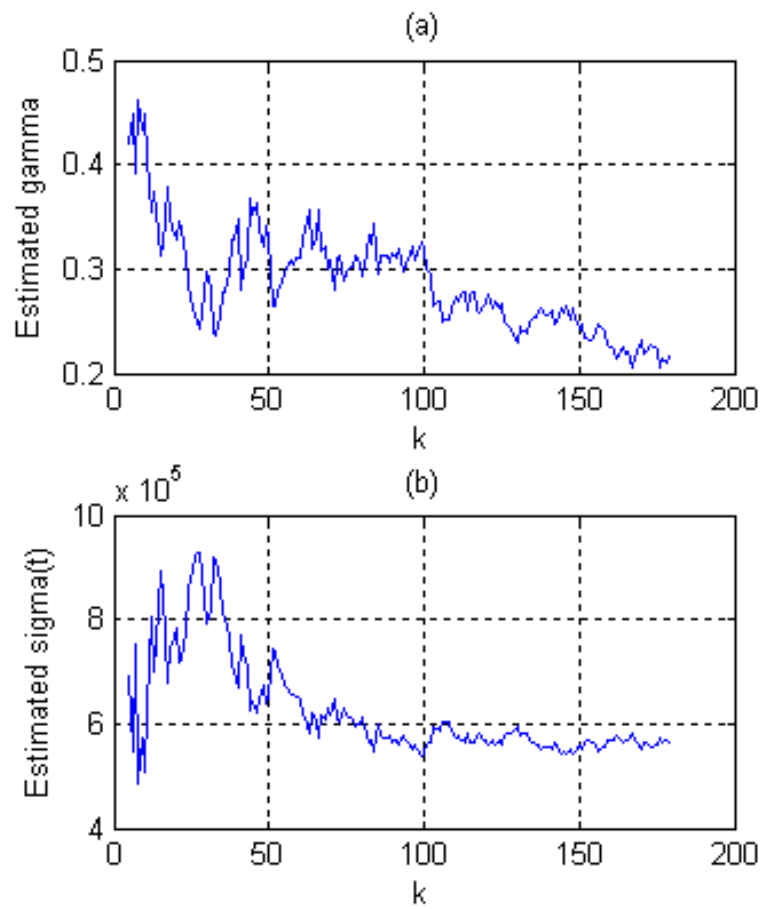


Figure 12. Estimates of γ and σ_t plotted against k

The estimate of the concentration parameter α as obtained through minimizing the negative differential entropy of the Dirichlet for $k = 100$,

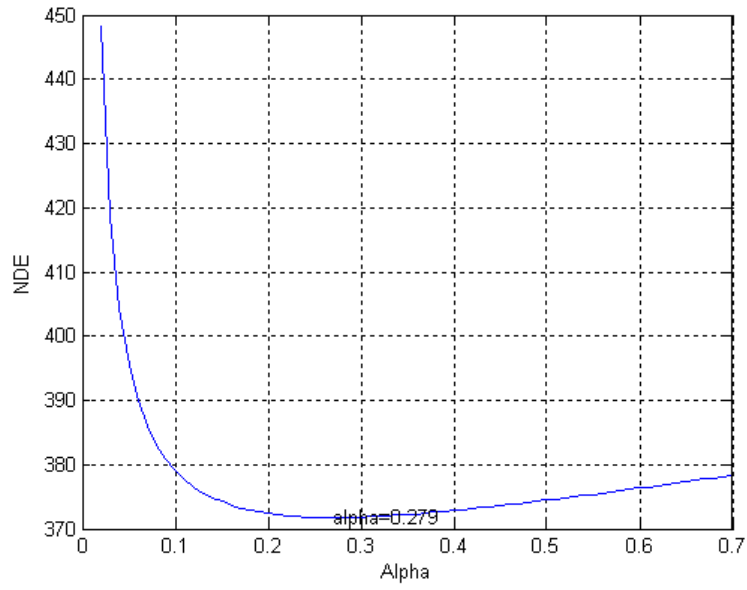


Figure 13. NDE at $k = 100$ indicating a minimum at $\alpha = 0.279$

After calculating the probabilities from H and G respectively that correspond to the empirical F_k , we can compare the three df's as shown in the probability plots in Figure 14.

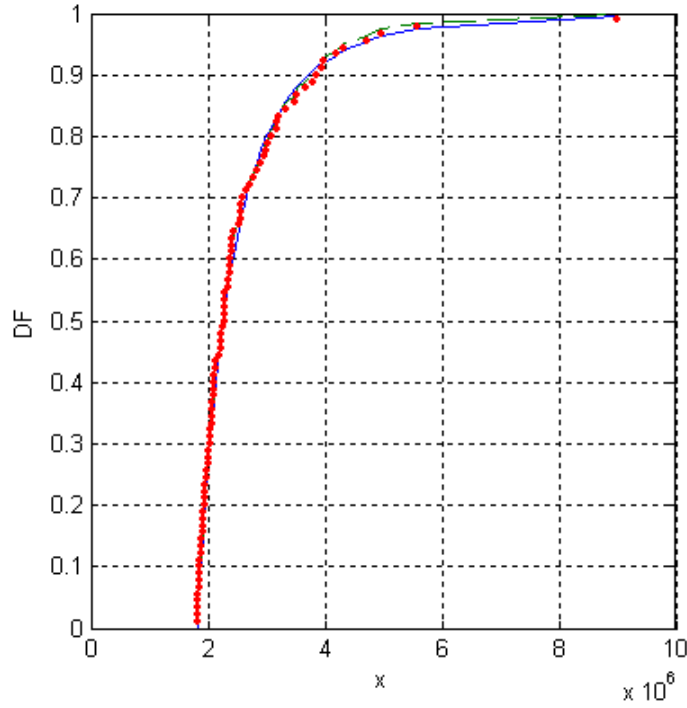


Figure 14. Distribution functions $G(x|t)$ (solid line), $H(x|t, \text{data})$ (broken line) and $F_k(x|t)$ (dots) calculated for the data exceeding t

From Figure 14, the fit of the GP seems to be a very good fit. The QQ -plots on the predictive quantiles calculated from the predictive distribution $H(x|t, \text{data})$ and the quantiles calculated from the GP $G(x|t)$ against the observed is, however, more instructive. Figure 15 shows such QQ -plots. The GP seems to be a good choice to describe the data. We notice, however, that the graphs start

to differ for high quantiles. This indicates that the prediction of high quantiles from the GP and from the predictive distribution may differ considerably.

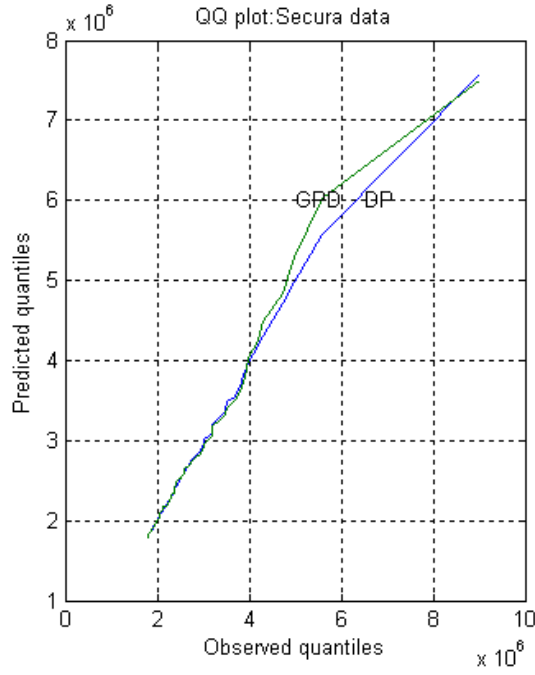


Figure 15. *QQ*-plots of GP quantiles (solid line) and predictive quantiles from $H(x|t, \text{data})$ (broken line) plotted against observed quantiles

5. Predicting high quantiles

The prediction of high quantiles exceeding the largest observation, can be done using the fact that the posterior distribution of $F(x|t)$ is $\text{Beta}(a_k, b_k)$. The mean is $\frac{a_k}{a_k + b_k} = H(x|t, \text{data})$, the predictive distribution. Let $\frac{a_k}{a_k + b_k}$ equal a given probability p , say 0.9994, then the quantile Q_p can be predicted. Figure 16 shows graphs of H and G beyond the largest observation 9.0037×10^6 exceeding the threshold t . The predicted

value is $\hat{Q}_p^H = 1.1320 \times 10^7$ for $a_k = 128.1021$ and $b_k = 0.0769$. The corresponding GP quantile is $\hat{Q}_p^G = 1.8384 \times 10^7$, which is much higher than the above obtained from H . Since $F(x|t)$ at $x = \hat{Q}_p^H$ is distributed $\text{Beta}(128.1021, 0.0769)$, the 95% hpd region for $F(\hat{Q}_p^H|t)$ can be calculated, namely (0.992, 0.999). This corresponds to the interval $(4.9594 \times 10^6, 9.6147 \times 10^6)$.

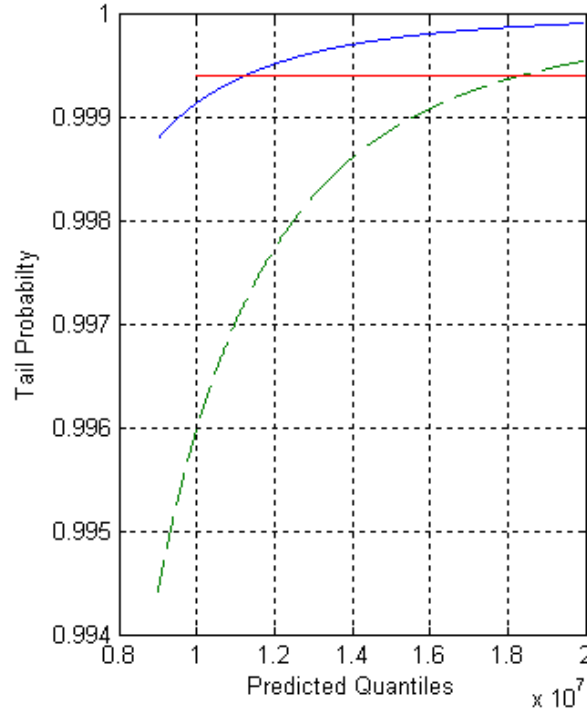


Figure 16. Graphs of H (thin line) and G (broken line) exceeding the largest observation indicating the $p = 0.9994^{\text{th}}$ line for predicting such a quantile

6. Conclusion

In conclusion one can ask what improvement is obtained by the proposed technique compared to existing techniques. We compared the choice of the threshold (or number of exceedances k) discussed here with the method of minimizing the asymptotic mean square error (see Beirlant et al., 2004, pp. 160–169) for the two data sets presented here, namely the Burr data and the Secura data. For the Burr data $k = 64$ is obtained and for the Secura data $k = 98$. This compares very closely to $k = 71$ and $k = 100$ respectively obtained through minimizing the relative NDE. The Hill estimates were respectively $\hat{\gamma}_{\text{Burr}} = 0.315$ and $\hat{\gamma}_{\text{Secura}} = 0.309$, both in the same region as that obtained through the proposed method. Confidence intervals can be calculated for the Hill estimates as shown by Beirlant et al., p. 162 which can be compared with the hpd regions which one can calculate from the posterior distributions of γ . The proposed method does not show real improvement, but merely gives the same results. However, it has the predictive ability with the possibility to calculate hpd regions from the predictive posterior distribution which the existing methods do not have.

The prediction of high quantiles considered here is an empirical Bayes approach with the opportunity to specify an underlying model for the data. The choice of the concentration parameter α and the threshold t follows from minimizing the entropy of the Dirichlet distribution where the parameters of the Dirichlet are based on the data, the concentration parameter α and the parameters of the proposed distribution G . The prediction of high quantiles through the predictive distribution derived through the DP, however, takes a fresh look at predicting these quantiles. The results on the DP and the NDE are given in an appendix.

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Appendix

Theorem 1 The NDE reaches a minimum if $\nu_i = 1$ for all $i = 1, \dots, k+1$ with $\min(J_{k+1}) = \log \Gamma(k+1)$ which gives a lower bound for the NDE and is convex in $k \geq 0$.

Proof Take derivatives with respect to ν_i , $i = 1, \dots, k+1$ in (5) using $\frac{d \log \Gamma(x)}{dx} = \Psi(x)$ and solve at zero for ν_i .

Definition 1 The Dirichlet Process

Partition the sample space according to the largest k ordered observations $x_1, \dots, x_k$ exceeding a threshold t . Let $F(x|t)$ be the true unknown df of the rv X and $G(x|t)$ be a proposed df with known parameters. Let $Y_1, \dots, Y_{k+1}$ be rv's where $Y_i = P(x_{i-1} < X < x_i|t) = F(x_i|t) - F(x_{i-1}|t)$, $i = 1, \dots, k$ and $Y_{k+1} = 1 - F(x_k|t)$. Let $Y_1, \dots, Y_{k+1}$ be jointly Dirichlet($\nu_1, \dots, \nu_{k+1}$) distributed with $\nu_i = \beta (G(x_i|t) - G(x_{i-1}|t))$, $i = 1, \dots, k$ and $\nu_{k+1} = \beta (1 - G(x_k|t))$ under the proposed model $G = E(F)$ as a prior of F . Then $F(x|t)$ is described through a Dirichlet Process with parameters $G(x|t)$ and β . This is denoted by $F(x|t) \sim DP(G(x|t), \beta)$.

Corollary 1 From definition 1 follows that a priori $F(x|t) \sim \text{Beta}(\beta G, \beta(1 - G))$ for any x an element of the sample space.

Corollary 2 β has to be proportional to $k+1$, say $\beta = \alpha(k+1)$.

Proof It is obvious that the NDE will be small if ν_i are equal, say α , for all i . Since $\sum_{j=1}^{k+1} \nu_j = \beta$ and the lower bound for the NDE is reached when $\nu_i = 1$, the Corollary follows.

Theorem 2 If $F(x|t) \sim DP(G(x|t), \beta)$ a priori, then a posteriori $F(x|t)|\text{data} \sim P(G(x|t) + (k+1)F_k(x|t), \beta + k)$.

Proof The likelihood function of the probabilities $y_1, \dots, y_{k+1}$ given the data, is given by

Like $(y_1, \dots, y_{k+1}|\text{data}) = \prod_{i=1}^{k+1} y_i$.

Since the prior distribution of the probabilities is Dirichlet with density function

$$\pi(y_1, \dots, y_{k+1}) \propto \prod_{i=1}^{k+1} y_i^{\nu_i-1}, y_i > 0, \sum_{i=1}^{k+1} y_i = 1,$$

the posterior is also Dirichlet with parameters $\nu_1^* = \nu_1 + (k+1)F_k(x_1)$, $\nu_2^* = \nu_2 + (k+1)\{F_k(x_2) - F_k(x_1)\}, \dots, \nu_{k+1}^* = \nu_{k+1} + (k+1)\{1 - F_k(x_k)\} = \nu_{k+1}$, where $F_k(x)$ is the empirical df defined as

$$F_k(x) = \frac{1}{k+1} \sum_{i=1}^k \delta_{x_i}(x).$$

Corollary 3 A posteriori $F(x|t)|\text{data} \sim \text{Beta}(\{\beta G + (k+1)F_k\}, \{\beta + k - \beta G - (k+1)F_k\})$

Proof The proof follows from corollary 1.

Theorem 3 Let $P(X > x) = \left(\frac{\eta}{\eta + x^\tau}\right)^\lambda$, $x > 0$, denotes the survival function of the Burr(η, τ, λ) distribution, then for t large, $P(X - t > x|X > t)^{t \uparrow \infty} \sim \left(1 + \gamma \frac{x}{\sigma(t)}\right)^{-1/\gamma}$, $x > 0$, which implies that the tail probability of the Burr can be approximated by that of the Generalized Pareto with parameters $\gamma^{-1} = \lambda\tau$ and $\sigma(t) = t\gamma(1 + \eta/t^\tau)$.

Proof

$$\begin{aligned} P(X - t > x|X > t) &= \frac{P(X > x+t)}{P(X > t)} \\ &= \left(\frac{\eta + t^\tau}{\eta + (x+t)^\tau}\right)^\lambda \\ &= \left(\frac{\left(1 + \frac{x}{t}\right)^\tau + \frac{\eta}{t^\tau}}{1 + \frac{\eta}{t^\tau}}\right)^{-\lambda} \\ &= \left(1 + \frac{\tau \frac{x}{t} + o(t^{-1})}{1 + \frac{\eta}{t^\tau}}\right)^{-\lambda} \\ &= \left(1 + \gamma \frac{x}{\gamma t(1 + \eta/t^\tau)}\right)^{-\lambda\tau} + o(1). \end{aligned}$$